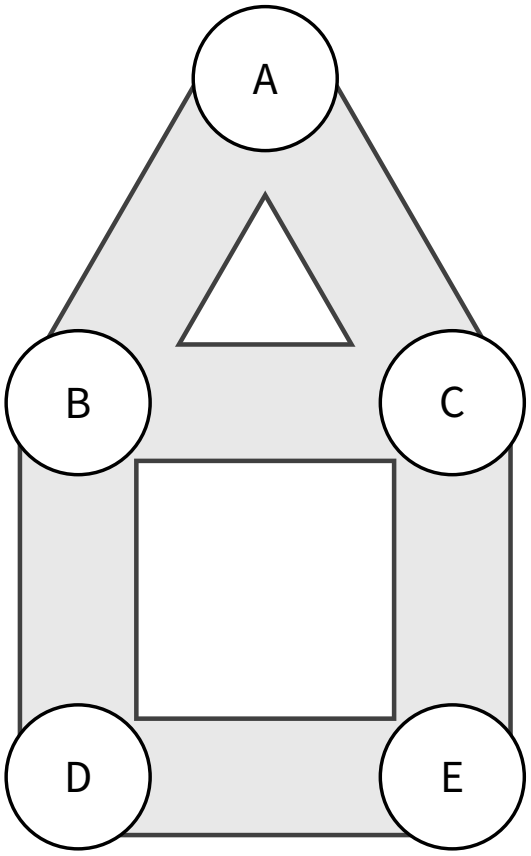
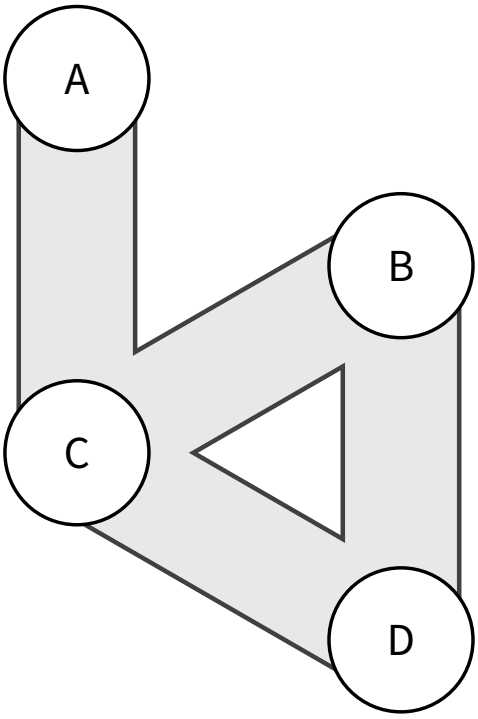
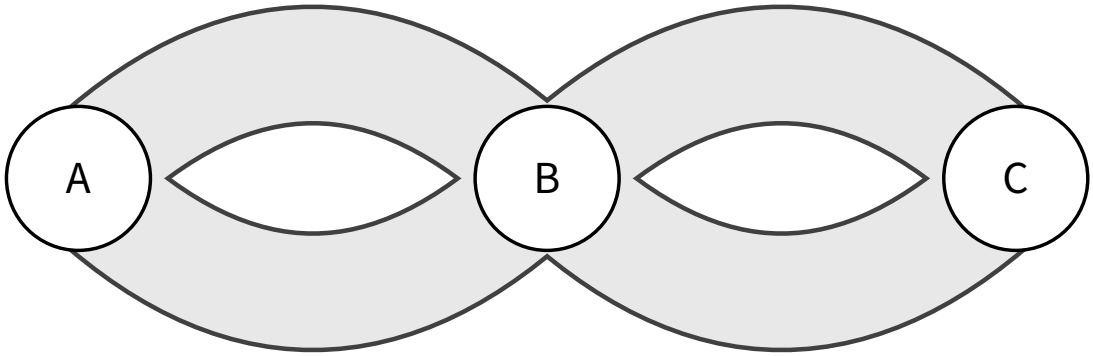
Put a counter on every bridge of a town before you walk it. A walk goes from island to island across bridges, and crossing a bridge means picking up its counter. A bridge whose counter is gone cannot be crossed again. Two islands can be joined by more than one bridge. In every town you can get from any island to any other.

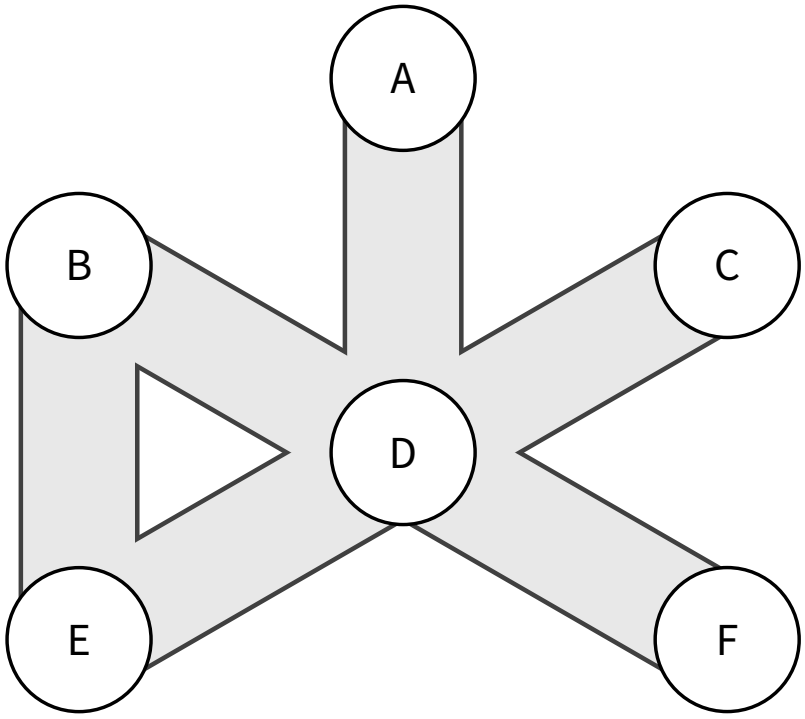
Problem 1: In each town, find a walk that crosses every bridge exactly once, and write its islands in order.

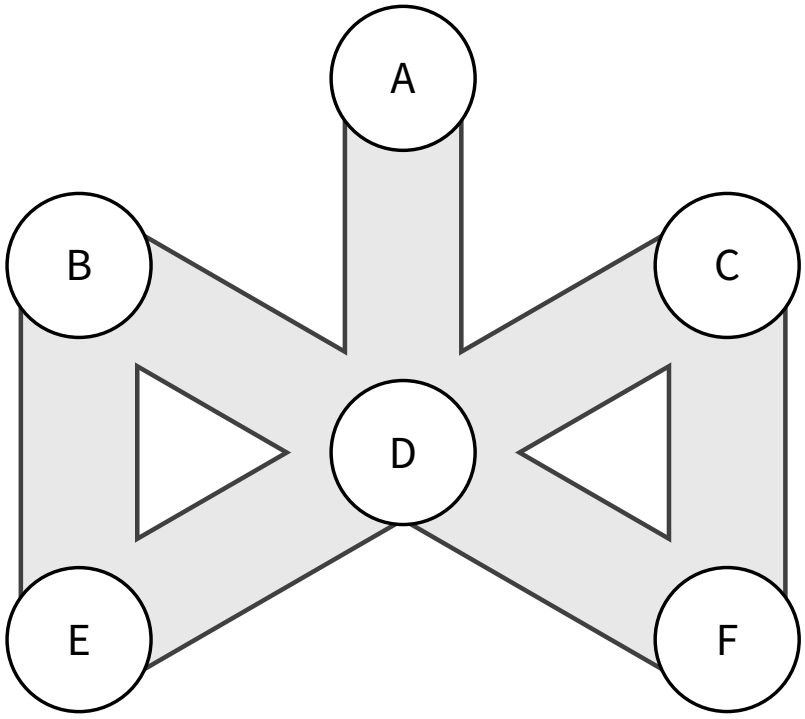


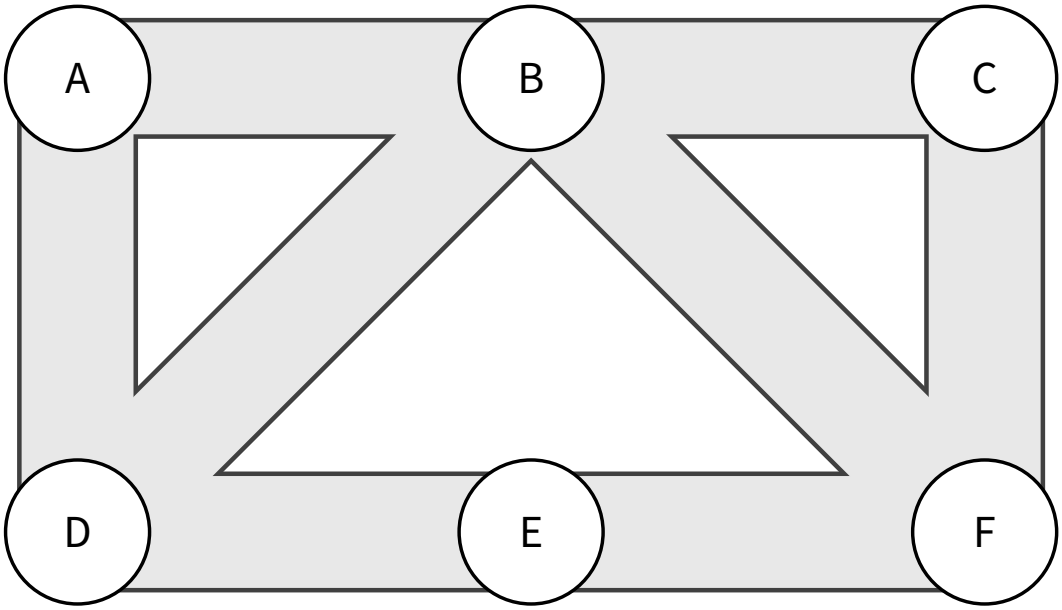


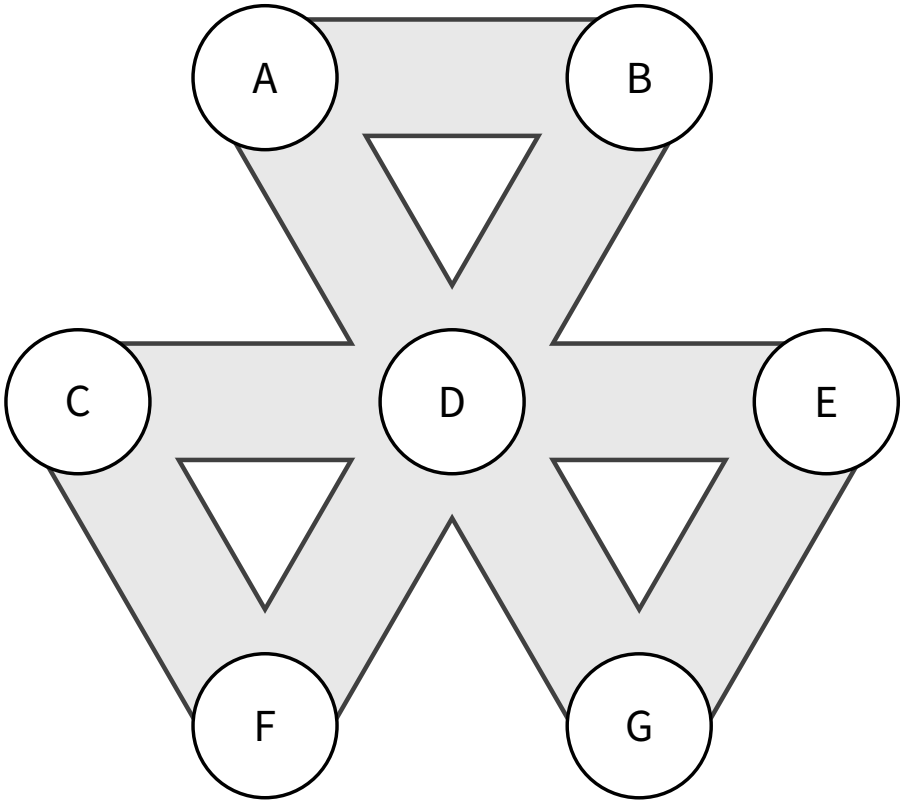


Problem 2: In each town, find every island where a walk that crosses every bridge exactly once can start, and where such a walk can end. If a town has no such walk, write “none”.

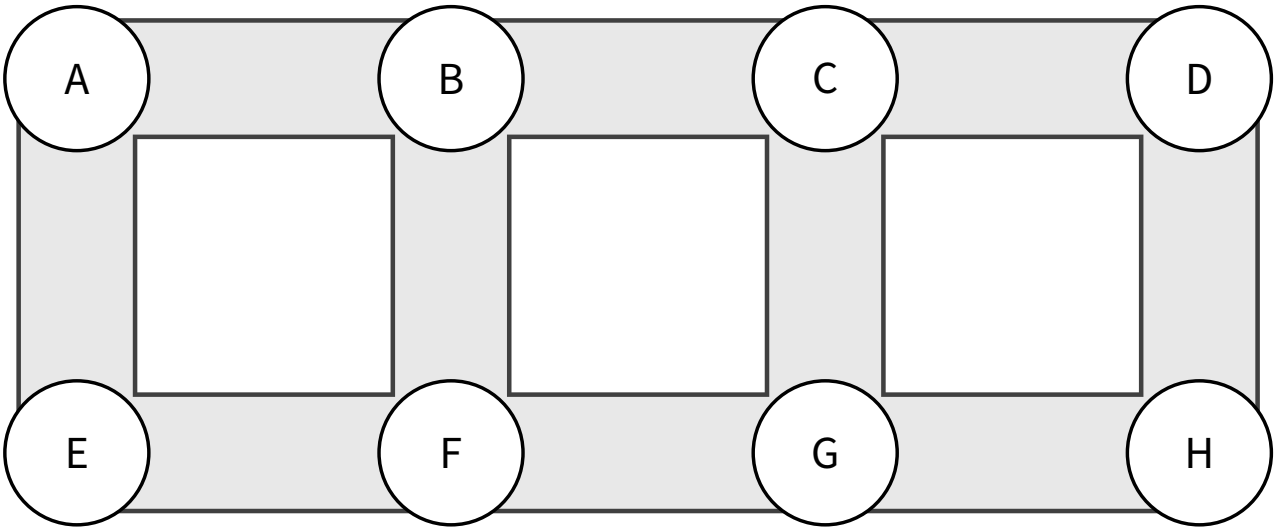


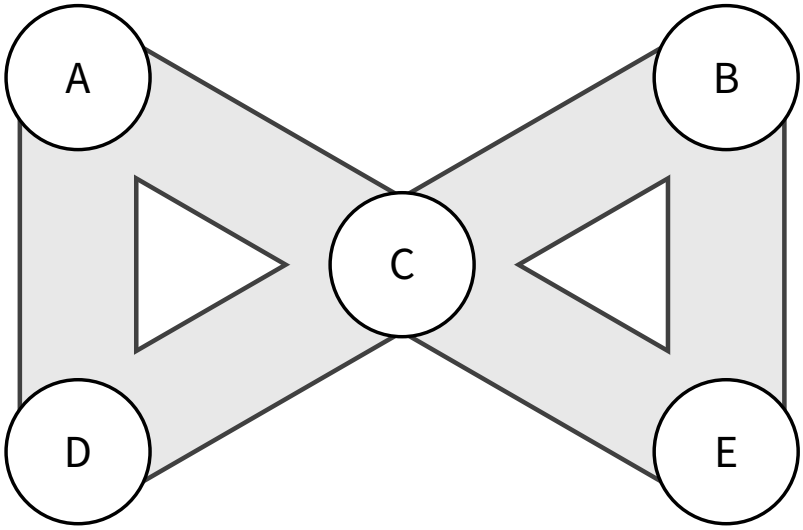


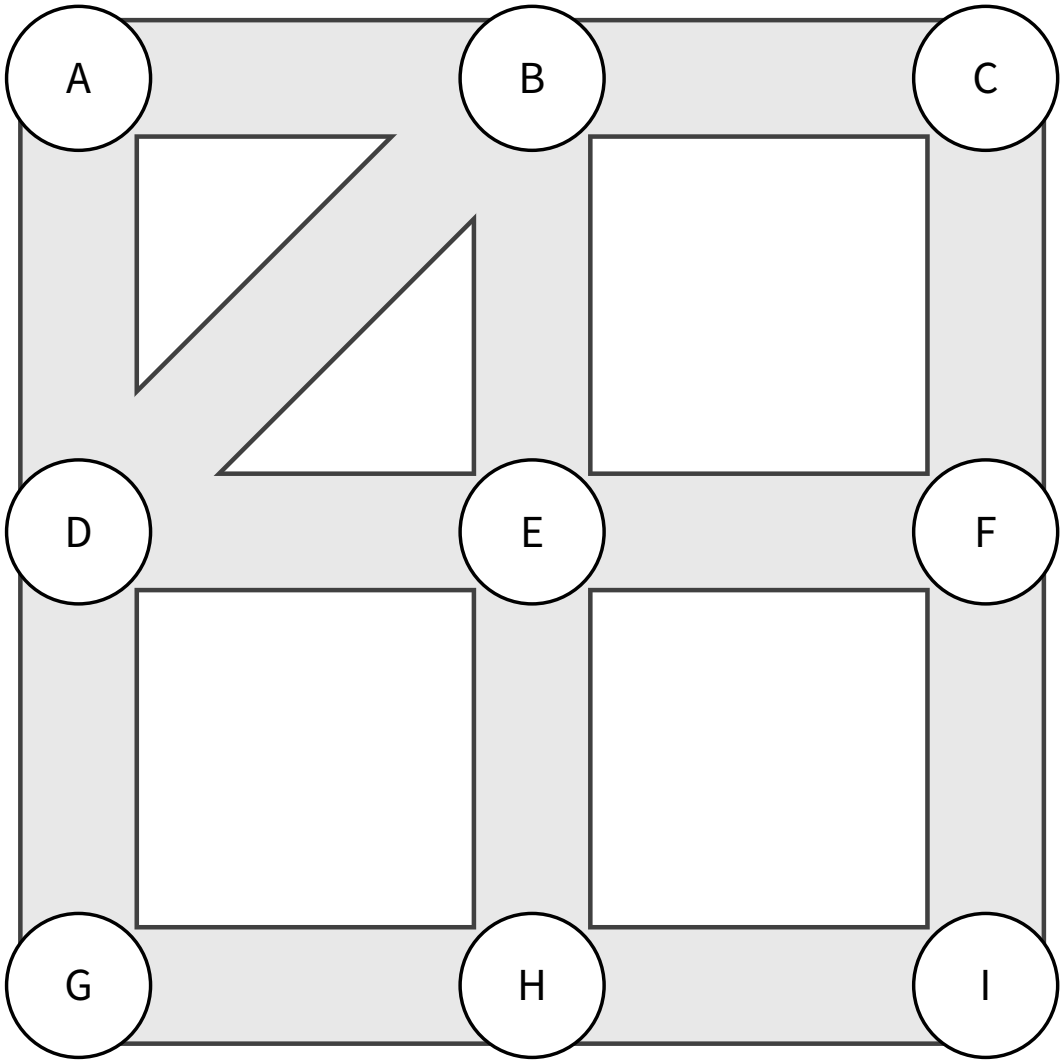

A
B
C
D
E
F


A
B
C
D
E
F
G

Problem 3: For each town, decide without walking it whether there is a walk that crosses every bridge exactly once, and if there is, where it can start and end. If there is none, write “none”. Then use counters to check each town that has a walk.







Problem 4: Write a rule that tells you, from a picture of a town, whether it has a walk that crosses every bridge exactly once, and where such a walk can start and end. Explain why a town that breaks your rule cannot have such a walk.

Problem 5: In the city of Königsberg, seven bridges joined four areas of land, A, B, C and D. Can someone walk through the city crossing every bridge exactly once? Explain. Where could the city build one new bridge so that such a walk becomes possible? What is the smallest number of new bridges the city needs for a walk that crosses every bridge exactly once and ends where it started?

